Split-explicit time integration for a mountain wave: MIS, RK3, exponential and Lawson methods with frozen and linearized fast parts

MultiRate.jl

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Abstract

We compare split-explicit time integrators for the compressible Euler equations on a terrain-following finite-volume grid, using the hydrostatic mountain wave over the Agnesi hill ($a = 10\,\mathrm{km}$) as test case. The fast part is integrated by a HEVI loop that couples horizontal Störmer–Verlet half steps with a vertically implicit Crank–Nicolson step. Four published MIS methods are compared with the Wicker–Skamarock RK3, two exponential (ETD) methods and a Lawson method, each for four ways of splitting the right-hand side: three *frozen* splittings (full pressure, frozen product form, extended system with the Exner pressure as an additional prognostic variable) and the classical *linearized* splitting $N = L_n + (N - L_n)$. Accuracy is measured against RK4 of the same semi-discrete operator, together with the largest stable time step and the dependence on the number of small steps.

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1 Introduction

In the compressible equations of atmospheric dynamics the fastest waves, sound waves with $c \approx 350\,\mathrm{m/s}$, carry almost no meteorologically relevant energy, while the flow of interest moves with 10–50 m/s. An explicit method for the whole system is limited by the acoustic Courant number and wastes most of its work on the acoustic modes. *Split-explicit* methods integrate the terms responsible for the sound waves with small time steps $\Delta\tau$ and the remaining ("slow") terms — advection, in our case also all nonlinear transport — with a large step Δt. The idea

goes back to Klemp and Wilhelmson (1978); the Runge–Kutta form of Wicker and Skamarock (2002), RK3 with forward–backward small steps, is the time integrator of WRF and many other models (Skamarock and Klemp 1992, 1994; Klemp, Skamarock and Dudhia 2007).

Multirate infinitesimal step (MIS) methods (Wensch, Knoth and Galant 2009; Knoth and Wensch 2014) generalize this construction. Each stage solves an ordinary differential equation for the fast part, whose initial value and forcing are linear combinations of the previous stages; the coefficients (α, β, γ) are free parameters that can be chosen for order and stability. RK3 is the MIS method with $\alpha = \gamma = 0$. Order conditions are derived assuming that the fast stages are solved exactly ("infinitesimal step"); in practice they are integrated with a few small steps. Four-stage MIS methods of order 3 (MIS4a) and five-stage methods with large stability regions (MIS54, Bauer and Knoth 2021) are available.

Exponential integrators (Hochbruck and Ostermann 2010) follow a related idea: the fast part L is integrated exactly, through the matrix functions $e^{\tau L}$ and $\varphi_k(\tau L)$, and the slow part enters as a polynomial forcing. In stage form ("exponential time differencing", ETD; Cox and Matthews 2002, Krogstad 2005) each stage is again a fast ODE with a forcing that is polynomial in time, and the φ-functions are replaced by a fast integrator; this makes ETD methods split-explicit methods of their own kind. Lawson methods (Lawson 1967) integrate in the frame of the fast flow (integrating factor). With exact fast stages and commuting fast and slow operators both reproduce the slow Runge–Kutta method on the slow part; in general they do not.

The splitting itself is a design decision. The classical approach linearizes the fast part about the state at the beginning of the large step, $N(q) = L_n q + (N(q) - L_n q)$, with a constant matrix L_n (*linearized* splitting). Alternatively the right-hand side can be written as $N(q) = F(q, q)$ with a first argument *frozen* at the stage values and a second one evolving in the fast integration (Knoth and Wensch, B-series and split-explicit methods); the fast part may then be nonlinear and its coefficients come from the stages, not from q^n. We compare three frozen forms: the full pressure $p(\Theta)$ in the fast part, the frozen product form $(\partial p/\partial\Theta)_f\,\delta\Theta$, and an *extended system* in which the Exner pressure π is an additional prognostic variable with its own (consistent) evolution equation, so that the pressure gradient is the product $c_p\Theta\nabla\pi$ with frozen Θ and the fast part is linear in the evolving variables.

The test case is the hydrostatic mountain wave of Gallus and Klemp (2000) on a terrain-following (Gal-Chen and Somerville 1975) grid. It contains the two features that make split-explicit integration delicate: vertically propagating gravity waves forced by the orography (slow, but with vertical structure on the grid scale) and the acoustic transients excited by the impulsive start and by the metric terms of the coordinate transformation.

2 Spatial discretization

Grid. Periodic channel $[0, L_x]$, model top H; Gal-Chen coordinate $z = z_s(x) + \eta J(x)$, $J = 1 - z_s/H$, cells uniform in (x, η); Arakawa C-grid: ρ, $\Theta = \rho\theta$ (and π) at the cell centres, $U = \rho u$ at the x-faces, $W = \rho w$ at the interior η-faces ($W = 0$ at the ground and the top is replaced by the kinematic condition through the contravariant flux).

Conservation laws. Mass and Θ in flux form,

$$\partial_t\rho = -\frac{1}{J}\Big[\delta_x(J_x U) + \delta_\eta\Omega\Big], \qquad \partial_t\Theta = -\frac{1}{J}\Big[\delta_x(\theta_x J_x U) + \delta_\eta(\theta_\eta\Omega)\Big], \qquad \Omega = W - z_x\bar{U},$$

$\bar{U}$ the four-point mean, $\Omega = 0$ at the ground and the top. The face values θ_x, θ_η are pseudo third-order upwind values (Jähn et al. 2015), the momentum advection uses the shift technique (Hicken et al. 2005; Jähn et al. 2015).

Pressure gradient in the metric form

$$-\partial_x p|_z = -\frac{1}{J_x}\Big[\delta_x(Jp) - \delta_\eta(z_x p_\sigma)\Big], \qquad -\partial_z p = -\frac{\delta_\eta p}{J\Delta\eta},$$

p_σ the mean at the η-faces, quadratically extrapolated at the ground and the top (so that the hydrostatic part has interior truncation error); gravity $-g\bar{\rho}$ at the W faces. No reference state is subtracted. A Rayleigh layer damps W above z_D.

Discrete balance. Each column is initialized in discrete hydrostatic balance of the W row of the respective form (rest-state residual $|N_W| \le 6 \cdot 10^{-13}$); the remaining $|N_U| \approx 2$–$4 \cdot 10^{-5}$ is the truncation error of the horizontal pressure gradient on the sloping coordinate surfaces.

3 Splittings and extended systems

The semi-discrete system is $q' = N(q)$, $q = (\rho, U, W, \Theta)$. All split-explicit methods need a fast part. We write $N(q) = F(q, q)$ with

$$F(y, z) = F_f(C(y), z) + S(y),$$

S the slow part (momentum advection, y only) and F_f the fast part, which depends on the evolving z and on coefficients $C(y)$ frozen from y, linearly in C (so that the MIS combination of stages is a combination of the C).

- **Frozen, full pressure** (`gradp`): $C = (\theta_x, \theta_\eta)$, the face values of θ; F_f contains the mass and Θ fluxes of z with frozen θ_f, the full pressure $p(\Theta_z)$ and gravity. Nonlinear in z through $p(\Theta)$; the vertical implicit step uses an approximate Jacobian.
- **Frozen product form** (`product`): additionally the face derivative $P_f = (\gamma p/\Theta)_f(y)$ is frozen, $-\partial_x p \approx -P_f\, \partial_x \Theta|_z$; F_f is linear in z, the vertical step is exactly Crank–Nicolson. On the terrain grid $N(q) = F(q, q)$ defines the operator (it differs from the full-pressure one by the product rule).
- **Extended system** (`exner`): the Exner pressure $\pi = (p/p_0)^{R/c_p}$ is an additional prognostic variable with

  $$\partial_t \pi = \frac{\partial \pi}{\partial \Theta}\partial_t \Theta, \qquad \nabla p = c_p \Theta \nabla \pi \quad \text{(exact)},$$

  frozen coefficients $\kappa_f = (\partial\pi/\partial\Theta)\theta_f$ per face and neighbour and the face mean $c_p\bar{\Theta}_f$ in the momentum rows. F_f is linear in $z = (\rho, U, W, \Theta, \pi)$. The system keeps $\pi = \pi(\Theta)$ in continuous time; after every large step π is reset from Θ.
- **Linearized** (`linear`): $N(q) = L_n q + (N(q) - L_n q)$, L_n the Jacobian-like fast operator frozen at q^n and refrozen every large step; the slow part is $N - L_n$ evaluated at the stages; N is the full-pressure operator.

4 Time integration schemes

MIS. With $Y_1 = q^n$, stage $i = 2, \dots, s+1$ solves over $\tau \in [0, d_i\Delta t]$, $d_i = \sum_j \beta_{ij}$,

$$Z' = F_f\Big(\sum_j w_{ij} C(Y_j), Z\Big) + \sum_j w_{ij} S(Y_j) + \sum_j \frac{\gamma_{ij}}{d_i \Delta t}(Y_j - q^n), \qquad Z(0) = q^n + \sum_j \alpha_{ij}(Y_j - q^n),$$

$w_{ij} = \beta_{ij}/d_i$, $Y_i = Z(d_i\Delta t)$, with $\lceil d_i n_{\mathrm{sub}} \rceil$ small steps. Published methods used here: MIS2 (2nd order), MIS3C, MIS4a (Knoth and Wensch 2014) and MIS54 (Bauer and Knoth 2021). With a frozen coefficient the order of MIS54 and MIS4a drops to 2 (order conditions of the bicoloured B-series); MISF53 and the application-specific MIS_FV53 restore order 3.

RK3 (Wicker–Skamarock). MIS with $\alpha = \gamma = 0$, $\beta = \mathrm{diag}(1/3, 1/2, 1)$: every stage restarts from q^n; order 2 for nonlinear fast parts.

ETD3, ETD4. Stage form of exponential RK methods: stage i integrates from q^n over $c_i\Delta t$

$$Z' = F_f(C(q^n), Z) + \sum_{j\le i}\sum_{k} P_{ijk}\,(\tau/c_i\Delta t)^{k-1}\,\tilde{S}(Y_j), \qquad \tilde{S}(Y) = N(Y) - F_f(C(q^n), Y),$$

with the polynomial coefficients P of the φ-function representation: ETD3 is the Cox–Matthews ETDRK3 ($c = \frac{1}{2}, 1, 1$), ETD4 Krogstad's ETDRK4 ($c = \frac{1}{2}, \frac{1}{2}, 1, 1$). The fast part is frozen at q^n (frozen forms) or is L_n (linearized).

Lawson4. Integrating-factor form of classical RK4: the increments $\Delta t\, a_{ij}\tilde{S}(Y_j)$ are propagated by the fast flow between the nodes $c_j \to c_i$; with a nonlinear fast flow the order drops to 2.

Number of small steps. For a stage of length $c\,\Delta t$, $\lceil c\, n_{\mathrm{sub}}\rceil$ small steps, $n_{\mathrm{sub}} = \lceil \Delta t/\Delta\tau_{\mathrm{ac}}\rceil$ (times a factor in Section 7), $\Delta\tau_{\mathrm{ac}} = 0.85\cdot 0.92\,\Delta x/c$ (4.3 s here). Fast work per large step: $\sum_i d_i$ (MIS), $\sum_i c_i$ (ETD), the node propagations (Lawson).

5 The fast integrator: Störmer–Verlet coupled with HEVI

The fast part is split into a horizontal part F_h (horizontal fluxes, the horizontal part $-z_x\bar{U}$ of Ω, the horizontal pressure gradient with its metric term, and all constant forcings) and a vertical part F_v (the W part of the η-flux, the vertical pressure gradient and gravity). One small step $\Delta\tau$ is the symmetric composition

$$\Phi_{\Delta\tau} = H_{\Delta\tau/2} \circ V_{\Delta\tau} \circ H_{\Delta\tau/2},$$

where H is a Störmer–Verlet step of the horizontal part (thermodynamic rows a quarter step, momentum a half step, thermodynamic rows a quarter step: position Verlet; Hairer, Lubich and Wanner 2006) and V the linearly implicit Crank–Nicolson step

$$z \leftarrow z + \left(I - \tfrac{\Delta\tau}{2} J_v\right)^{-1} \Delta\tau\, F_v(z)$$

of the vertical part, solved column by column: ρ, Θ (and π) are eliminated, leaving a tridiagonal system for W (as in WRF). For the linear fast parts (product form, extended system) V is exactly Crank–Nicolson. The composition is symmetric and of second order; the horizontal part is limited by the horizontal acoustic Courant number, the vertical one is unconditionally stable, which removes the restriction $\Delta\tau < \Delta z/c$ ($\Delta z/c \approx 0.6$ s here, against $\Delta\tau_{\mathrm{ac}} = 4.3$ s). No divergence damping (extrapolation) is used in this study.

6 Test case

Agnesi hill $h(x) = h_0/(1 + (x - x_m)^2/a^2)$, $h_0 = 400$ m, $a = 10$ km; $U = 10$ m/s, $N = 0.01\,\mathrm{s}^{-1}$, $\theta_0 = 300$ K ($Na/U = 10$: hydrostatic regime, Gallus and Klemp 2000); channel $40a = 400$ km, top 15.6 km, Rayleigh damping of W above 9 km (rate $5c_{gz}/(H - z_D)$, $c_{gz} = U\min(1, U/Na)$); $\Delta x = 2$ km (200 columns), $\Delta\eta = 200$ m (78 layers); impulsive start; integration to $T = 10.8\,a/U = 10800$ s. Reference: classical RK4 of the same semi-discrete operator with $\Delta t = 0.21$ s (vertical acoustic Courant number 0.4).

Error measure. Relative L^2 error of the potential temperature perturbation $\theta' = \theta - \bar{\theta}(z)$ in the box $|x - x_m| \le 3a$, $z \le 9\,\mathrm{km}$ (below the damping layer, where the steady wave dominates). The error of w is reported as well, but it is not a useful accuracy measure here: the impulsive start excites vertical acoustic modes that fill the whole periodic channel and are not damped (the Rayleigh layer acts only above 9 km). They carry vertical velocity but almost no θ'. RK4 with $\Delta t = 0.21\,\mathrm{s}$ transports them correctly. The split methods, whose vertical step is Crank–Nicolson with vertical acoustic Courant number $c\Delta\tau/\Delta z \approx 8$, cannot represent their phase, so every split run differs from RK4 in w by 7–19 %, whatever Δt and n_{sub} (Figure 2). The difference does not converge: MIS2 at $\Delta t = 10\,\mathrm{s}$ gives 0.070 with $n_{\mathrm{sub}} = 3$ and 0.16 with $n_{\mathrm{sub}} = 12$ or 48, while the θ' error falls from 0.016 to 0.0022 and 0.0019. The same difference appears between two split methods (MIS2 against MIS3C at $\Delta t = 10\,\mathrm{s}$: 8 %). It is not a sensitivity of the flow: a relative perturbation of 10^{-8} of the initial Θ changes the MIS2 errors in the fourth digit only.

7 Results

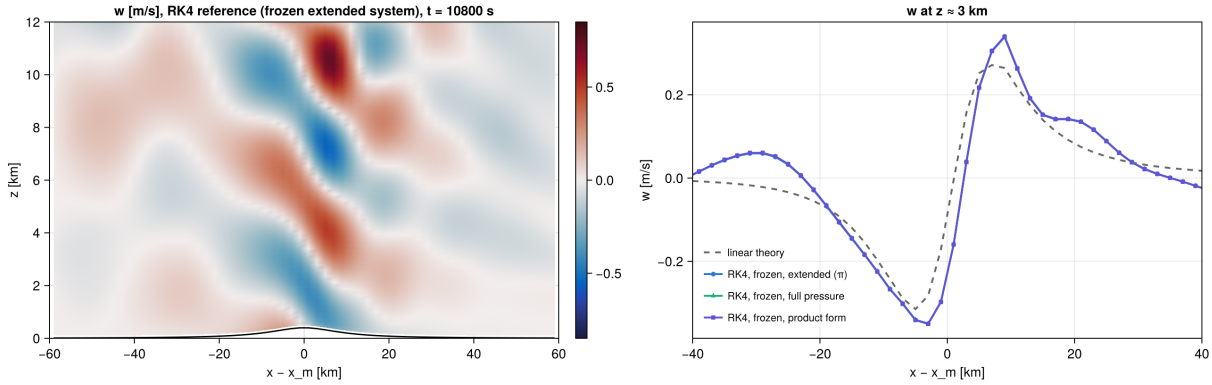


Figure 1: Reference solution: w at $t = 10800\,\mathrm{s}$ (RK4, extended system), and w at $z \approx 3\,\mathrm{km}$ for the three frozen operators and linear theory.

Reference. The three RK4 references (one per operator; the linearized splitting has the operator of the full-pressure form) give the same wave, $\max|w| = 0.847$, 0.849, 0.849 m/s (Figure 1). At $z = 3\,\mathrm{km}$ the steady linear solution is reproduced in amplitude and phase upstream and over the hill; the deviation in the lee is the not yet stationary part of the wave and the transients of the impulsive start. The three operators agree to plotting accuracy, so the differences in this study are caused by the time integration only.

Small time steps. With $n_{\mathrm{sub}} = \lceil \Delta t/\Delta\tau_{\mathrm{ac}} \rceil$ the small step is 3.3–4.3 s for every Δt; for small Δt the error is that of the fast integrator. All stable methods give a θ' error of 1.2–1.9 % at $\Delta t = 10\,\mathrm{s}$, Lawson4 0.7 %. It falls with $\Delta\tau$: MIS2 0.016 ($\Delta t = 10\,\mathrm{s}$, $\Delta\tau = 3.3\,\mathrm{s}$), 0.0071 ($\Delta t = 5\,\mathrm{s}$, $\Delta\tau = 2.5\,\mathrm{s}$), 0.0017 ($\Delta t = \Delta\tau = 2.5\,\mathrm{s}$); MIS3C at $\Delta t = 5\,\mathrm{s}$ 0.0045 and, with four times as many small steps, 0.0008.

Errors. Relative L^2 error of θ' ($n_{\mathrm{sub}} = \lceil \Delta t/4.3\,\mathrm{s} \rceil$; u: unstable, i.e. non-finite or $\max|w| > 17\,\mathrm{m/s}$ within T):

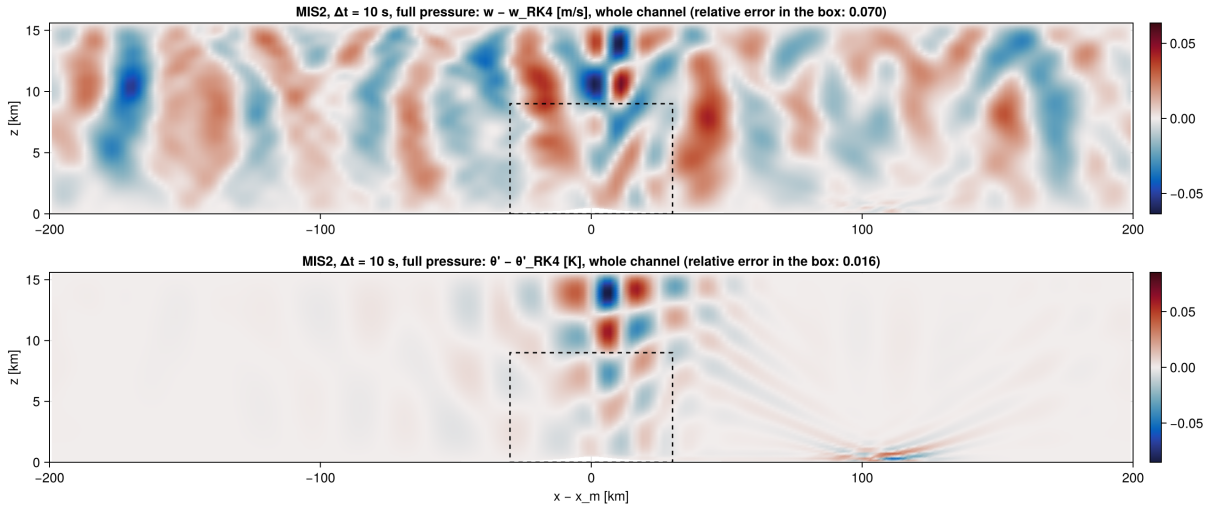


Figure 2: MIS2, $\Delta t = 10\,\mathrm{s}$, full pressure, minus RK4 in the whole channel. Top: w, dominated by the undamped vertical acoustic modes of the impulsive start; bottom: θ', the phase error of the mountain wave above the hill. Dashed: the error box.

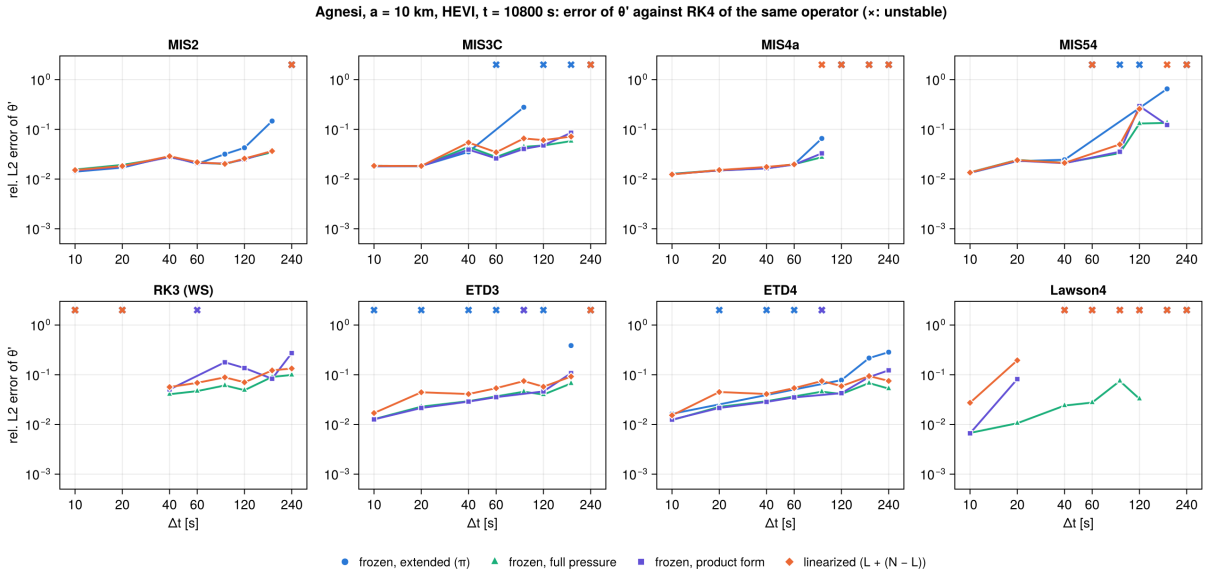


Figure 3: Relative L^2 error of θ' against Δt, one panel per method, one line per splitting: the three frozen splittings in blue, green and violet, the linearized splitting in orange; ×: unstable. $n_{\mathrm{sub}} = \lceil \Delta t / 4.3\,\mathrm{s} \rceil$.

Δt [s]	10	20	40	60	90	120	180	240
$C_{\text{sound}} = c\Delta t/\Delta x$	1.8	3.6	7.3	10.9	16.4	21.8	32.7	43.7
$C_{\text{adv}} = U\Delta t/\Delta x$	0.05	0.1	0.2	0.3	0.45	0.6	0.9	1.2
frozen, extended (π)								
MIS2	0.014	0.017	0.029	0.020	0.032	0.043	0.147	u
MIS3C	0.018	0.018	0.035	u	0.278	u	u	u
MIS4a	0.013	0.015	0.016	0.020	0.066	u	u	u
MIS54	0.013	0.023	0.024	u	u	u	0.652	u
RK3 (WS)	u	u	u	u	u	u	u	u
ETD3	u	u	u	u	u	u	0.386	u
ETD4	0.016	u	u	u	u	0.078	0.217	0.285
Lawson4	u	u	u	u	u	u	u	u
frozen, full pressure								
MIS2	0.016	0.019	0.028	0.022	0.020	0.025	0.035	u
MIS3C	0.019	0.018	0.044	0.028	0.045	0.047	0.058	u
MIS4a	0.013	0.015	0.017	0.020	0.028	u	u	u
MIS54	0.014	0.024	0.021	u	0.033	0.131	0.135	u
RK3 (WS)	u	u	0.041	0.047	0.061	0.049	0.090	0.099
ETD3	0.013	0.023	0.029	0.037	0.046	0.040	0.067	u
ETD4	0.012	0.023	0.029	0.037	0.046	0.041	0.068	0.053
Lawson4	0.007	0.011	0.024	0.028	0.075	0.033	u	u
frozen, product form								
MIS2	0.015	0.018	0.028	0.021	0.021	0.025	0.037	u
MIS3C	0.018	0.018	0.039	0.026	0.040	0.048	0.085	u
MIS4a	0.013	0.015	0.017	0.020	0.033	u	u	u
MIS54	0.013	0.023	0.021	u	0.035	0.294	0.123	u
RK3 (WS)	u	u	0.051	u	0.178	0.136	0.083	0.273
ETD3	0.013	0.021	0.029	0.035	u	0.046	0.108	u
ETD4	0.013	0.021	0.028	0.035	u	0.043	0.090	0.123
Lawson4	0.007	0.082	u	u	u	u	u	u
linearized ($L + (N - L)$)								
MIS2	0.015	0.018	0.029	0.022	0.020	0.026	0.037	u
MIS3C	0.018	0.018	0.054	0.035	0.066	0.061	0.072	u
MIS4a	0.012	0.015	0.018	0.020	u	u	u	u
MIS54	0.014	0.024	0.021	u	0.050	0.258	u	u
RK3 (WS)	u	u	0.057	0.069	0.089	0.070	0.122	0.134
ETD3	0.017	0.044	0.041	0.054	0.075	0.057	0.092	u
ETD4	0.015	0.045	0.041	0.054	0.075	0.059	0.094	0.075
Lawson4	0.027	0.195	u	u	u	u	u	u

The error grows only slowly with Δt: MIS2 goes from 1.6 % at $\Delta t = 10$ s to 3.5 % at 180 s for the full-pressure form, i.e. at fixed $\Delta\tau$ the large step contributes little to the error. In the stable range the three p-based splittings (full pressure, product form, linearized) give practically the same MIS errors; the differences are in the stability. The frozen and linearized splittings differ most for the methods whose fast stages start from q^n and use a fast part frozen at q^n (ETD, Lawson, RK3): the frozen full-pressure form is about 30 % more accurate than the linearized one for RK3, ETD3 and ETD4 (e.g. ETD4, $\Delta t = 60$ s: 0.037 against 0.054). The reason is that the frozen full-pressure fast part evaluates $p(\Theta)$ along the fast trajectory, while L_n linearizes it at q^n.

Large time step. Largest Δt [s] with a θ' error below 10 % ("–": none):

	frozen, π	frozen, p	frozen, product	linearized
MIS2	120	180	180	180
MIS3C	40	180	180	180
MIS4a	90	90	90	60
MIS54	40	90	90	90
RK3 (WS)	–	240	180	120
ETD3	–	180	120	180
ETD4	120	240	180	240
Lawson4	–	120	20	10

Observations:

- **MIS2 and MIS3C** are stable and accurate up to $\Delta t = 180\,\mathrm{s}$ ($C_{\mathrm{sound}} = 33$, $C_{\mathrm{adv}} = 0.9$) in all three p-based splittings, and fail at 240 s ($C_{\mathrm{adv}} = 1.2$), presumably the advective limit of the slow part with third-order upwind fluxes.

- **MIS4a** is the most accurate MIS method below its limit but fails from $\Delta t = 120\,\mathrm{s}$ on ($C_{\mathrm{sound}} \geq 22$) in every splitting, in the linearized one already at 90 s. This matches the Knoth–Wensch stability chart of MIS4a, whose stable advective range collapses for large sound Courant numbers.

- **MIS54** has gaps: unstable at $\Delta t = 60\,\mathrm{s}$ in every splitting, stable at 90 s, inaccurate (13–29 %) from 120 s on. The instability at 60 s persists with 2, 4 and 8 times as many small steps (full pressure and extended system), so it belongs to the method with (nearly) exact fast stages, not to the fast integrator: a resonance of the stage lengths $d_i \Delta t$ with the gravity-wave and acoustic frequencies of the discrete system. The error field at 120 s (Figure 4) is a regular, nearly monochromatic pattern, a weakly growing mode.

- **RK3 (WS)** is unstable at $\Delta t = 10$ and 20 s ($C_{\mathrm{adv}} \leq 0.1$) and stable from 40 s on, up to 240 s. This is the behaviour predicted by the stability theory: with $\alpha = \gamma = 0$ the acoustic amplification factor has $|g_0| = 1$, and the second-order term grows for small C_{adv} (without divergence damping, which is not used here). At large Δt the advective damping of the third-order upwind slow part suffices. Its error is 1.5–3 times the MIS2 error at the same Δt.

- **ETD3, ETD4** are stable in the p-based splittings up to 180–240 s with errors between MIS2 and RK3, at 2.1–2.5 times the fast work of MIS2. In the product form they fail at $\Delta t = 90\,\mathrm{s}$ (a gap similar to MIS54).

- **Lawson4** works only with the full nonlinear fast part (full pressure) up to 120 s; with the product form or the linearized fast part it fails from $\Delta t = 40\,\mathrm{s}$ on. In the Lawson method the increments $\Delta t\, a_{ij} \tilde{S}(Y_j)$ are propagated by the fast flow between the nodes; with a fast part linear in z that is not the linearization of N along the solution, the propagated increments do not cancel, and the defect is amplified by the fast flow.

- **The extended system** (frozen, π) is the least robust: RK3, ETD3 and Lawson4 are unstable at almost every Δt, ETD4 at 20–90 s, MIS3C and MIS54 from 60 s on; only MIS2 (and MIS4a up to its own limit) are as robust as for the p-based forms. See the discussion.

The error fields at $\Delta t = 120\,\mathrm{s}$ (Figure 4) show the three kinds of error. MIS2 has a smooth phase error of the wave above the hill. RK3 and ETD4 have errors of similar size, but small-scale and layered in the vertical, everywhere in the domain: modes that are poorly resolved by the fast integrator and are not damped, because these methods have $|g_0| = 1$. MIS54 shows a regular growing mode.

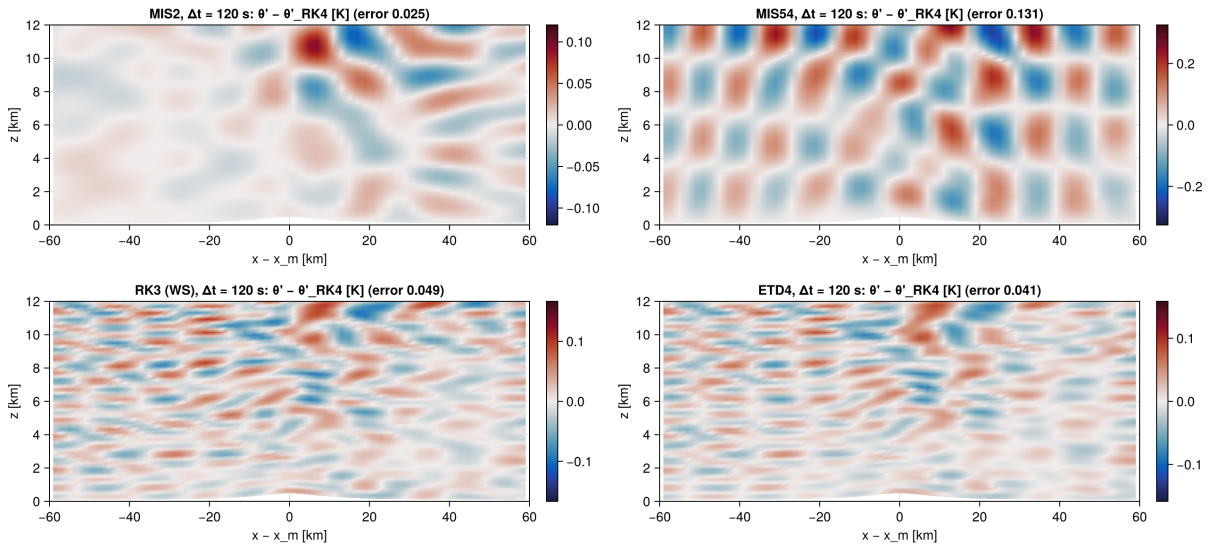


Figure 4: $\theta' - \theta'_{\mathrm{RK4}}$ at $\Delta t = 120\,\mathrm{s}$ (frozen, full pressure). MIS2: smooth phase error of the wave; MIS54: a growing regular mode near the stability limit; RK3 and ETD4: undamped small-scale noise layered in the vertical.

Number of small steps. $\Delta t = 60\,\mathrm{s}$, $n_{\mathrm{sub}} = 14\,f$ ($\Delta\tau = 4.3\,\mathrm{s}/f$), θ' error:

f	1	2	4	8
frozen, extended (π)				
MIS2	0.0203	0.0068	0.0053	0.0044
MIS3C	u	u	u	u
MIS4a	0.0198	u	u	u
MIS54	u	u	u	u
RK3 (WS)	u	u	u	u
ETD3	u	u	u	u
ETD4	u	u	u	u
Lawson4	u	u	u	u
frozen, full pressure				
MIS2	0.0216	0.0073	0.0057	0.0049
MIS3C	0.0277	0.0155	0.0147	0.0151
MIS4a	0.0199	u	u	u
MIS54	u	u	u	u
RK3 (WS)	0.0469	0.0368	0.0387	0.0318
ETD3	0.0367	0.0283	0.0128	0.0221
ETD4	0.0367	0.0283	0.0125	0.0221
Lawson4	0.0276	0.0380	0.0192	0.0293

With the standard number of small steps ($f = 1$, $\Delta\tau = 4.3\,\mathrm{s}$) the error is dominated by the fast integrator. MIS2 drops from 0.022 to 0.0073 when the small step is halved and then saturates at 0.005 (extended system 0.0044). That is the error of the large step $\Delta t = 60\,\mathrm{s}$ itself, about a quarter of the total error at $f = 1$. MIS3C saturates at 0.015, a larger large-step error than MIS2. For RK3, ETD3, ETD4 and Lawson4 the error does not decrease monotonically (e.g. ETD4: 0.037, 0.028, 0.013, 0.022). These methods do not damp the poorly resolved fast modes ($|g_0| = 1$), and the phase of these modes, and with it the error, depends on n_{sub}.

MIS4a and MIS54 behave in the opposite way to the intuition "more small steps, more accuracy": MIS4a is stable with $f = 1$ and unstable with $f = 2, 4, 8$, and MIS54 is unstable for every f. With few small steps the fast integrator (horizontal Störmer–Verlet at acoustic Courant number 0.78, Crank–Nicolson in the vertical) adds damping and dispersion to the high fast frequencies.

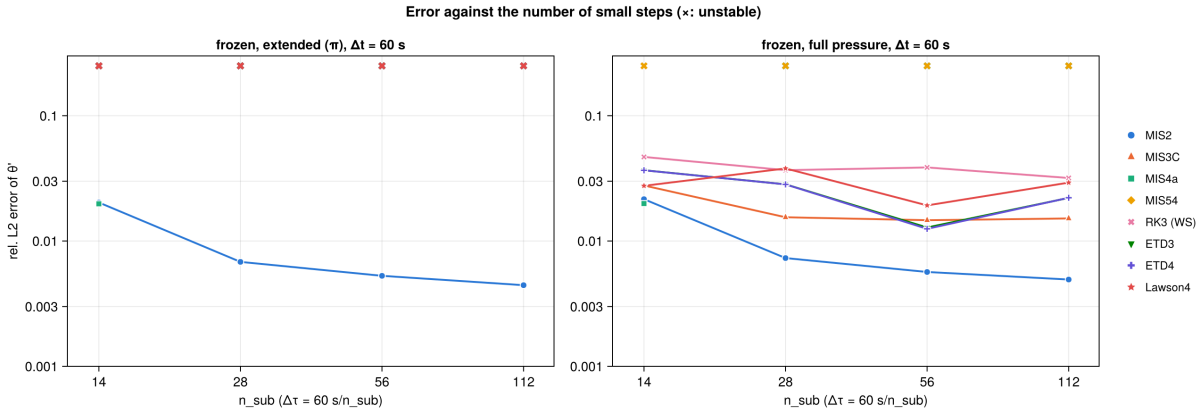


Figure 5: θ' error against the number of small steps, $\Delta t = 60\,\mathrm{s}$; left: extended system, right: full pressure; $\times$: unstable.

With more small steps the fast stages approach the exact flow, and $\Delta t = 60\,\mathrm{s}$ ($C_{\mathrm{sound}} = 11$) lies outside the stability region of MIS4a and MIS54 with exact fast stages. This agrees with the observation in the stability study that, at fixed divergence damping, more small steps weaken the damping of the method. The stability of a split-explicit method is therefore a property of the pair (MIS method, fast integrator with its $\Delta\tau$), and a method should be qualified with the n_{sub} it is used with. The extended system with $f \geq 2$ is stable only with MIS2.

8 Discussion

Why MIS is robust. The stability theory of the companion report gives the amplification factor of a pure acoustic mode as $R(i\omega, \mu) = e^{i\omega} \sum_k g_k \mu^k$, μ the slow eigenvalue. For RK3, ETD and Lawson $|g_0| = 1$: an acoustic or poorly resolved fast mode is neither damped nor amplified to leading order, and the next term decides. For RK3 this term amplifies for small advective Courant numbers, which is why RK3 is unstable at $\Delta t \leq 20\,\mathrm{s}$ here and why WRF needs divergence damping. MIS methods with $\alpha, \gamma \neq 0$ have $|g_0| < 1$ (MIS2, MIS3C, MIS4a: $1 - |g_0| \sim \omega^4$, MIS54: ω^6), a built-in damping of the fast modes that is independent of the fast integrator. The test confirms both sides: without any divergence damping MIS2 and MIS3C are stable from $\Delta t = 10$ to $180\,\mathrm{s}$ in all p-based splittings, and the error fields at large Δt are smooth for MIS2, but contain undamped small-scale noise for RK3 and ETD4 (Figure 4).

Accuracy and cost. With $n_{\mathrm{sub}} \propto \Delta t$ the small step is fixed, and the error is dominated by the fast integrator. The large step adds little: the θ' error of MIS2 doubles from $\Delta t = 10$ to $180\,\mathrm{s}$. The cost is the fast work, $\sum_i d_i\, n_{\mathrm{sub}}$ small steps per large step: over the run 2991 small steps for MIS2 and MIS3C, 3709 for MIS4a, 4620 for RK3, 4732 for MIS54, 6300 for ETD3 and Lawson4, 7560 for ETD4 (each with three HEVI column solves). MIS2 has the lowest fast work, its error is within 10 % of the best MIS method (MIS4a) where both are stable, and only RK3 and ETD4 reach a larger step. At $\Delta t = 60\,\mathrm{s}$, ETD4 needs 2.5 times the fast work of MIS2 for 1.7 times its error, and RK3 needs 1.5 times the work for 2.1 times the error.

The higher-order MIS methods. MIS4a and MIS54 are not better than MIS2 here. Their higher order (reduced to 2 by the frozen coefficients anyway) is invisible behind the fast-integrator error. Their stability regions have gaps at large sound Courant numbers (MIS4a from $C_{\mathrm{sound}} \approx 20$, MIS54 at $C_{\mathrm{sound}} \approx 11$), which are properties of the methods with exact fast stages: more small steps make them worse (Section 7). The idealized Knoth–Wensch test predicts such limits, but

not their exact positions, which depend on the frequency content of the discrete operator (gravity waves, terrain metric).

The splitting. For MIS methods the splitting hardly matters, as long as p is the prognostic pressure: full pressure, product form and linearized splitting give the same errors and the same stability limits within one Δt step of the scan. For the methods that start every stage from q^n with a fast part frozen at q^n (RK3, ETD, Lawson) the frozen full-pressure form is clearly best. It is about 30 % more accurate than the linearized one and is the only form in which Lawson4 works beyond $\Delta t = 20\,\mathrm{s}$. The product form is a poor compromise for these methods (RK3 unstable at 60 s, ETD at 90 s, Lawson4 from 40 s).

The extended system. The extended system with the Exner pressure is less robust than the p-based forms for every method except MIS2. The cause is not the fast part: the spectra of the frozen fast operators of the three frozen forms coincide on a reduced grid (40×26 cells, same hill and Δx, $\Delta\eta$): $\max \Re\lambda = 5.3 \cdot 10^{-5}\,\mathrm{s}^{-1}$ on the terrain grid (machine zero on flat ground), and the same imaginary parts. The difference is in the full operator N: linearized about the initial flow over the hill, the extended system has real eigenvalues up to $1.4 \cdot 10^{-3}\,\mathrm{s}^{-1}$, against $4.8 \cdot 10^{-7}\,\mathrm{s}^{-1}$ for the p-based operators. The eigenvectors are pure constraint defects: a perturbation $\delta\Theta$ without the consistent $\delta\pi = \pi_\Theta \delta\Theta$ ($|\delta\pi - \pi_\Theta \delta\Theta| = 31\,|\delta\pi|$), with negligible momentum, located in the lowest layer on the upslope side. They are absent on flat ground and at $U = 0$. The extended system carries an additional degree of freedom, the defect $\pi - \pi(\Theta)$, which is neutral in the continuous equations but is not preserved by the discrete π equation in the sloping lowest layer, where the contravariant flux converges. The reset $\pi \leftarrow \pi(\Theta)$ after each large step does not cure the instability (the same runs are unstable without it), so the defect is created and amplified within the step, by the stage combinations of the frozen coefficients. The growth rate alone ($\lambda \Delta t \approx 0.08$ at $\Delta t = 60\,\mathrm{s}$) does not explain the blow-up; we therefore regard the defect mode as the mechanism to be removed, not as a complete explanation. Possible remedies are a π equation that uses exactly the discrete divergence of the Θ equation multiplied by π_Θ of the same cell, a relaxation of π towards $\pi(\Theta)$ inside the fast loop, or the full-pressure frozen form, which has no additional variable.

Recommendation for this configuration. MIS2 (or MIS3C) with the frozen full-pressure splitting and HEVI Störmer–Verlet/Crank–Nicolson small steps at $\Delta\tau \approx 0.8\,\Delta x/c$: stable without divergence damping from $\Delta t = 10$ to 180 s (C_{sound} up to 33) with a θ' error of 2–4 % against RK4, at the lowest fast work of all methods tested. The accuracy is limited by the small step, not by Δt. The vertical acoustic modes of the impulsive start remain unresolved in phase by every split method; this concerns w, not the mountain wave.

A Error of w

Relative L^2 error of w in the same box. It is dominated by the vertical acoustic modes of the impulsive start (Figure 2) and therefore does not decrease with Δt:

Δt [s]	10	20	40	60	90	120	180	240
frozen, extended (π)								
MIS2	0.114	0.105	0.116	0.109	0.122	0.205	1.292	u
MIS3C	0.074	0.139	0.139	u	16.602	u	u	u
MIS4a	0.181	0.093	0.098	0.102	2.912	u	u	u
MIS54	0.167	0.085	1.356	u	u	u	2.070	u
RK3 (WS)	u	u	u	u	u	u	u	u
ETD3	u	u	u	u	u	u	1.671	u
ETD4	1.220	u	u	u	u	2.146	0.777	1.259
Lawson4	u	u	u	u	u	u	u	u
frozen, full pressure								
MIS2	0.070	0.105	0.112	0.106	0.110	0.112	0.111	u
MIS3C	0.087	0.156	0.121	0.104	0.123	0.131	0.148	u
MIS4a	0.178	0.105	0.098	0.101	1.522	u	u	u
MIS54	0.178	0.093	0.088	u	0.133	1.275	0.330	u
RK3 (WS)	u	u	0.142	0.159	0.204	0.128	0.134	0.162
ETD3	0.206	0.140	0.122	0.175	0.146	0.191	0.123	u
ETD4	0.200	0.132	0.097	0.148	0.139	0.186	0.123	0.136
Lawson4	0.166	0.181	0.115	0.106	0.146	0.231	u	u
frozen, product form								
MIS2	0.086	0.106	0.112	0.107	0.101	0.128	0.171	u
MIS3C	0.084	0.151	0.116	0.111	0.116	0.138	0.454	u
MIS4a	0.175	0.100	0.098	0.101	1.861	u	u	u
MIS54	0.173	0.091	0.083	u	0.131	3.485	0.255	u
RK3 (WS)	u	u	4.297	u	1.683	1.594	0.223	1.296
ETD3	0.245	0.305	1.367	0.447	u	0.270	0.561	u
ETD4	0.214	0.147	0.237	0.393	u	0.266	0.350	0.645
Lawson4	0.207	2.239	u	u	u	u	u	u
linearized ($L + (N - L)$)								
MIS2	0.071	0.105	0.114	0.107	0.111	0.112	0.113	u
MIS3C	0.087	0.157	0.130	0.110	0.144	0.145	0.159	u
MIS4a	0.179	0.106	0.098	0.101	u	u	u	u
MIS54	0.179	0.094	0.088	u	0.187	2.794	u	u
RK3 (WS)	u	u	0.183	0.199	0.267	0.187	0.188	0.223
ETD3	0.265	0.165	0.142	0.206	0.201	0.245	0.181	u
ETD4	0.246	0.152	0.116	0.179	0.191	0.239	0.181	0.185
Lawson4	0.167	0.660	u	u	u	u	u	u

References

[1] S. Bauer, O. Knoth: Extended multirate infinitesimal step methods: derivation of order conditions. J. Comput. Appl. Math. 387 (2021), 112541.

[2] S.M. Cox, P.C. Matthews: Exponential time differencing for stiff systems. J. Comput. Phys. 176 (2002), 430–455.

[3] T. Gal-Chen, R.C.J. Somerville: On the use of a coordinate transformation for the solution of the Navier–Stokes equations. J. Comput. Phys. 17 (1975), 209–228.

[4] W.A. Gallus, J.B. Klemp: Behavior of flow over step orography. Mon. Wea. Rev. 128 (2000), 1153–1164.

[5] E. Hairer, C. Lubich, G. Wanner: Geometric Numerical Integration, 2nd ed. Springer 2006.

[6] J.E. Hicken, F.E. Ham, J. Militzer, M. Koksal: A shift transformation for fully conservative methods: turbulence simulation on complex, unstructured grids. J. Comput. Phys. 208 (2005), 704–734.

[7] M. Hochbruck, A. Ostermann: Exponential integrators. Acta Numerica 19 (2010), 209–286.

[8] M. Jähn, O. Knoth, M. König, U. Vogelsberg: ASAM v2.7: a compressible atmospheric model with a Cartesian cut cell approach. Geosci. Model Dev. 8 (2015), 317–340.

[9] J.B. Klemp, R.B. Wilhelmson: The simulation of three-dimensional convective storm dynamics. J. Atmos. Sci. 35 (1978), 1070–1096.

[10] J.B. Klemp, W.C. Skamarock, J. Dudhia: Conservative split-explicit time integration methods for the compressible nonhydrostatic equations. Mon. Wea. Rev. 135 (2007), 2897–2913.

[11] J.B. Klemp, W.C. Skamarock, S.-H. Ha: Damping acoustic modes in compressible horizontally explicit vertically implicit (HEVI) and split-explicit time integration schemes. Mon. Wea. Rev. 146 (2018), 1911–1923.

[12] O. Knoth, J. Wensch: Generalized split-explicit Runge–Kutta methods for the compressible Euler equations. Mon. Wea. Rev. 142 (2014), 2067–2081.

[13] O. Knoth, J. Wensch: B-series and split-explicit methods. Manuscript, `https://github.com/OsKnoth/B-Series`.

[14] S. Krogstad: Generalized integrating factor methods for stiff PDEs. J. Comput. Phys. 203 (2005), 72–88.

[15] J.D. Lawson: Generalized Runge–Kutta processes for stable systems with large Lipschitz constants. SIAM J. Numer. Anal. 4 (1967), 372–380.

[16] W.C. Skamarock, J.B. Klemp: The stability of time-split numerical methods for the hydrostatic and the nonhydrostatic elastic equations. Mon. Wea. Rev. 120 (1992), 2109–2127.

[17] W.C. Skamarock, J.B. Klemp: Efficiency and accuracy of the Klemp–Wilhelmson time-splitting technique. Mon. Wea. Rev. 122 (1994), 2623–2630.

[18] J. Wensch, O. Knoth, A. Galant: Multirate infinitesimal step methods for atmospheric flow simulation. BIT Numer. Math. 49 (2009), 449–473.

[19] L.J. Wicker, W.C. Skamarock: Time-splitting methods for elastic models using forward time schemes. Mon. Wea. Rev. 130 (2002), 2088–2097.